

Producing algorithms: flowcharts

Revise 360

Worksheet for the 360° experience: Flowchart design studio

Name: _____

Class: _____

Date: _____

How to use this sheet

1. Discuss or research the Do Now on slide 1. Complete the starter and key terminology below.
2. Open Flowchart design studio. Read each station panel and write the key facts in your own words. Complete its worksheet tasks before tapping the badge.
3. Answer the questions on screen. Your first answer counts. Use review mode to learn from mistakes.
4. Copy your score into the yellow box, then complete the exam practice from memory.

By the end of the lesson I will:

- Recognise flowchart symbols.
- Follow decision paths.
- Create a complete flowchart.

Starter: before you put the headset on

A program reads age, checks age ≥ 12 and prints a result. Which step is a decision?

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Key terminology

Flowchart

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Decision

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The stations

Stations 1 and 2: turn right. Stations 3 and 4: turn around. Stations 5 and 6: turn left. Final challenge: look down.

1 Start and finish

Turn right

Key facts in your own words:

Example to trace:

TERMINAL: start / end

ARROW: next step

Follow one path at a time

Challenge: Why should an arrow have a clear direction?

Match the symbol name to its role.

Match each item to a description from the list below.

Calculation or assignment / Direction of execution / Start or end

Item	Matching description
Terminal	
Flow line	
Process	

Which symbol marks the end?

2 Read and write

Turn right

Key facts in your own words:

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Example to trace:

INPUT: borrower ID
INPUT: item ID
OUTPUT: loan result

Challenge: Which symbol should contain Input age?

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Which action belongs in an input/output symbol?

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Output total is drawn in...

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3 Process the data

Turn around

Key facts in your own words:

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Example to trace:

PROCESS
charge = hours * rate
total = total + charge

Challenge: Which symbol should contain total = total + 3?

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Which label is a process?

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Assign total = 0 is drawn in...

.....

4 Branch on a decision

Turn around

Key facts in your own words:

Example to trace:

DECISION: available?
Yes: record loan
No: show unavailable
Both paths reach End

Challenge: What is missing if only the Yes path is drawn?

What belongs inside a decision diamond?

The two paths from a condition can be labelled...

5 Use a subprogram

Turn left

Key facts in your own words:

Example to trace:

MAIN FLOW
Read item ID
Call checkAvailability
Use returned result

Challenge: How does a subprogram symbol help a large flowchart?

A rectangle with extra vertical lines at its sides usually means...

A subprogram symbol refers to...

6 Check every route

Turn left

Key facts in your own words:

Example to trace:

Test available = true
Expected: loan recorded
Test available = false
Expected: unavailable

Challenge: Why test both branches?

Order a simple flowchart design process.

Put these steps in order: Trace each possible branch / Connect and label the paths / Choose symbols for the steps / State inputs and required outputs

A disconnected branch means...

★ Final challenge (look down)

Before you tap the star, complete these tasks on paper.

Choose a flowchart symbol for each action.

Categories: *Input/output / Process / Decision*

Item	Category
Read hours	
charge = hours * 3	
charge > 12?	
Print charge	

What must a decision have in this two-way flowchart?

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Your score

Copy it from the score screen as soon as it appears.

_____ / 22

Key question

How could a diagram make a decision easier to check?

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Review and improve

Use review mode. Explain one correction and why the correct answer fits. Create the flowchart in a diagram tool or on paper. Implement it in Python and check ages 11, 12 and 13.

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Exam practice

Near the end of the lesson, close the experience and answer from memory.

a) Name the symbols for input, a calculation and a decision. [3]

b) Draw a flowchart that inputs age, outputs Allowed if age ≥ 12 and otherwise outputs Too young. [5]

c) Explain why testing only one path is insufficient. [2]

How confident am I?

Objective	Not yet	Getting there	Confident
Recognise flowchart symbols.			
Follow decision paths.			
Create a complete flowchart.			